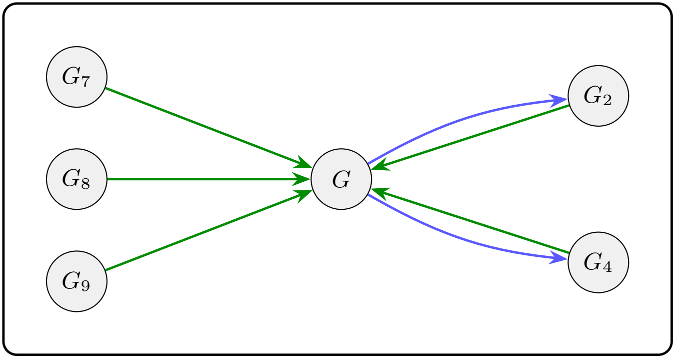
graphs$_2$ (Box I)



Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box I).

1. $G \xrightarrow{x_1 \to \bar{x}_4\, x_1} G_2$ *(Step 2)*.
2. $G \xrightarrow{\bar{x}_4 \to x_1\, \bar{x}_4} G_4$ *(Step 2)*.
3. $G_2 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_1,\ x_2 \mapsto \bar{x}_3,\ x_3 \mapsto \bar{x}_2,\ x_4 \mapsto \bar{x}_5,\ x_5 \mapsto \bar{x}_4$ *(permutation / ep-isomorphism)*.
4. $G_4 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5,\ x_2 \mapsto x_3,\ x_3 \mapsto \bar{x}_4,\ x_4 \mapsto \bar{x}_1,\ x_5 \mapsto \bar{x}_2$ *(permutation / ep-isomorphism)*.
5. $G_7 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_1,\ x_2 \mapsto x_2,\ x_3 \mapsto x_3,\ x_4 \mapsto x_5,\ x_5 \mapsto x_4$ *(permutation / ep-isomorphism)*.
6. $G_8 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5,\ x_2 \mapsto x_4,\ x_3 \mapsto \bar{x}_3,\ x_4 \mapsto x_1,\ x_5 \mapsto x_2$ *(permutation / ep-isomorphism)*.
7. $G_9 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5,\ x_2 \mapsto x_4,\ x_3 \mapsto \bar{x}_3,\ x_4 \mapsto x_2,\ x_5 \mapsto x_1$ *(permutation / ep-isomorphism)*.